



DEALER BEST PRACTICE SERIES

3-Stage Wash Bay For Cold Climate

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC
Management

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1.0 Introduction

This document is an example of a practice undertaken at a specific work-site. The detail provided may be site specific therefore the information within should be taken as a suggestion. Please note that the intention of this best practice is to provide an 'idea' for reducing delays due to the washing of equipment in cold weather. Specific design detail has not been provided and may need to be further investigated depending on site requirements.

The best practice has been created as a suggestion for the following:

- Efficient use of water
- Proper after treatment and disposal of contaminated water and mud
- Reduced washing time during freezing period

During winter, temperatures at this site average -23 C (-10F), although can drop as low as -40 C (-40F). The time taken to wash a production truck can be more than eight hours due to the ice build-up caused by spraying water on frozen surfaces. This situation can bring about increased planning and shop coordination for PM and specific repairs. This **eight-hour** process coupled with the occupancy of the washing area can cause delays to other machines. Implementation of a warm-up area can result in increased production. During winter, the warm-up and primary wash processes are the most time consuming steps, both approximately equal in duration. With the use of only one bay initially, it can take about four hours to defrost and four hours to clean equipment.

The combination of oil and soap produces a compound that is expensive to treat. Therefore, there is a need to separate the washing cycle into two processes to first remove oil and to later use soap to clean. This also assists to reduce water consumption.

The two processes are as follows:

- Primary
- Finishing

Note that soap is only used during the 'finishing' process, with minimal water volume.

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2.0 Best Practice Description

The main project consists of three areas:

1. Warm-up area
2. Primary wash area
3. Finishing area

They are complemented with three facilities:

1. Component washing area
2. Mud collecting area
3. Water-oil treatment plant

A separate automated wash bay is used for washing light vehicles and service trucks. This shares the same setup, such as high pressure jets and a hydraulic lift, to allow equipment to be effectively cleaned.

Warm-up area: This is the means of entry into the wash process. The warm up area is separated from the wash bay by a vertical door. High capacity forced air heaters direct the hot airflow to the machine. Soap-free water is used to clean the floor with a high-pressure hose. The facility also contains a soap-free high-pressure water and steam. Mud is directed to a mud-pit by water pressure jets located in a gutter. The gutters are located throughout the midsection of the bay and on the entry and exit sides of each bay. Large amounts of mud are moved with a skid steer loader. Grease is hand shoveled into a used-grease tub, located in the bay, for future treatment. To allow year-round efficient utilization, this area has an adjacent bay on one side for the tire shop. See the layout plan in section six for further clarification. This allows the warm-up area to be used for maneuvering the tire manipulator or as an additional bay for quick fixes when the main shop is crowded. *On average, it now takes two hours to defrost equipment.*

Primary wash area: This bay breaks down mud and oils which are directed to a mud pit via gutters. A simple oil-water separator exists to separate the oil, which would otherwise not be possible if soap were used. Also, a larger volume of water is required at this stage resulting in a larger volume of runoff into the gutters. *On average, this process takes one and a half hours.*

This area is equipped with nine soap-free water canons. Components, such as shovel masts and track frames, can be washed in this area whilst onboard a trailer before proceeding to the component washing area. The following list is a configuration of the setup:

- Each side of the primary wash area contains two fixed canons on an elevated catwalk
- Two sets of canons are at floor level. One set on each side of the wash area.
- One mobile water canon
- Soap-free water/steam high pressure washer

Finishing area: With the oil and mud mostly cleaned from the machine, high pressure steam and soap cleaning equipment can now be used to clean the machine with a low volume of water. The

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emulsified water and oil is directed to a sump and is pumped to a treatment plant. *On average, this process takes one and a half hours.*

Component washing area: This area is adjacent to the machine wash bay. See the layout plan in section six for further clarification. The following list is a configuration of the setup:

- Fifteen ton overhead crane
- High pressure steam and soap cleaning equipment
- Emulsified water and oil is directed to a sump and is pumped to a treatment plant.

A floor cleaning hose is not used in this area to minimize the water volume being processed at the water-oil treatment plant.

Mud settling basin: The finishing and component wash areas use water jets to direct dirty water and mud to the mud settling basin through gutters. Note that emulsified water and oil are not directed to the mud settling basin. A skid steer loader assists with larger volumes of mud. A gravity fed water-oil separator transfers any overflow from the basin to an exterior basin for final treatment. A wheel loader is used to transfer mud to an external storage area once the mud-settling basin becomes full.

Water-oil treatment plant- This is a specialized unit that makes use of physical and chemical processes to separate the emulsified oils from the water. Note that specific details for the treatment plant have not been made available.

3.0 Implementation Steps

Evaluate:

- Physical space requirements for current and future equipment
- Utilization time of washing installation
- Water consumption needed for each area
- Mud quantities to be transferred into the holding basin and treatment areas Volume of emulsified water and oil to be treated and treatment areas
- Secondary usage of warm-up area when not in use

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4.0 Benefits

- Average time in each bay
 - o Defrost: 2 hours
 - o Primary 1.5 hours
 - o Finishing 1.5 hours
- Overall time saving: 37.5% (average)
 - o Reduction of warm-up time by 50%
 - o Reduction of total wash time by 30%
- Improved worker efficiency. Reduced wash-bay delays improve worker effectiveness.
- Improved overall efficiency and quality of the washing.
- Reduction of non-productive time through an improvement of the wash-bay effectiveness, mainly due to the machine being in the warm-up area instead of waiting outside.
- Reduction in the amount of emulsified water and oil to be treated
- Ability to contain more equipment in the overall process.

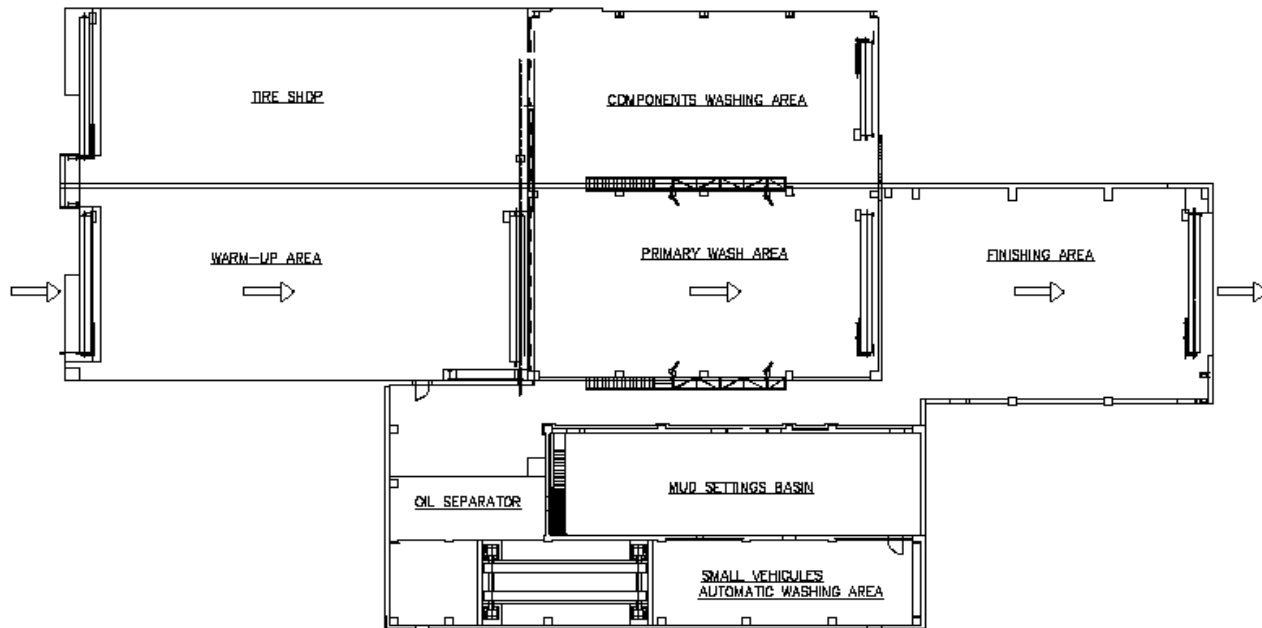
5.0 Resources Required

- Engineering department to develop the project
- Management acceptance for funding this major investment
- Training – Two people are required to operate this facility

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6.0 Supporting Attachments / References

3-STAGE WASH BAY FOR COLD CLIMATE Mont-Wright, Quebec.



7.0 Related Best Practices

8.0 Acknowledgements

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